REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	ECN00239 - APPROVED 11/19/2020	01/27/2021	A. NICHOLS

UL Color Codes

Abbreviation	Color
AL	Aluminum
ALL	Any color
AM	Amber
AO	All opaque
AT	All transparent
BG	Beige
BK	Black
BL	Blue
BN	Brown
BZ	Bronze
CH	Charcoal
CL	Clear
CT	Clear transparent
DK	Dark
GD	Gold
GN	Green
GY	Gray
GT	Granite
IV	Ivory
LT	Light
NC	Natural (no pigmentation)
OL	Olive
OP	Opaque
OR	Orange
PK	Pink
RD	Red
SM	Smoke
TL	Translucent
TN	Tan
TP	Transparent
VT	Violet
WT	White
YL	Yellow

Unless otherwise noted, the following specifications must be followed.

Wire Specifications:

- **Mil-Spec:** MIL-W-16878/4
- **RoHS:** Not Required

Example: Belden 83007

P: Power Cable
R: RF Cable
C: Control/Command/Data
M: Mixed

PROPRIETARY AND CONFIDENTIAL

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UNCLASSIFIED

-	-	DRAWN BY	DATE
-	-	S. MANNING	10/15/2019
-	-	CHECKED BY	DATE
-	-	C. BURGIN	10/16/2019
-	-	ENGINEER	DATE
-	-	C. BURGIN	10/16/2019
-	-	APPROVED BY	DATE
ASY00478	YELLOWSTONE	A. NICHOLS	10/16/2019
NEXT ASSY	USED ON		
APPLICATION			

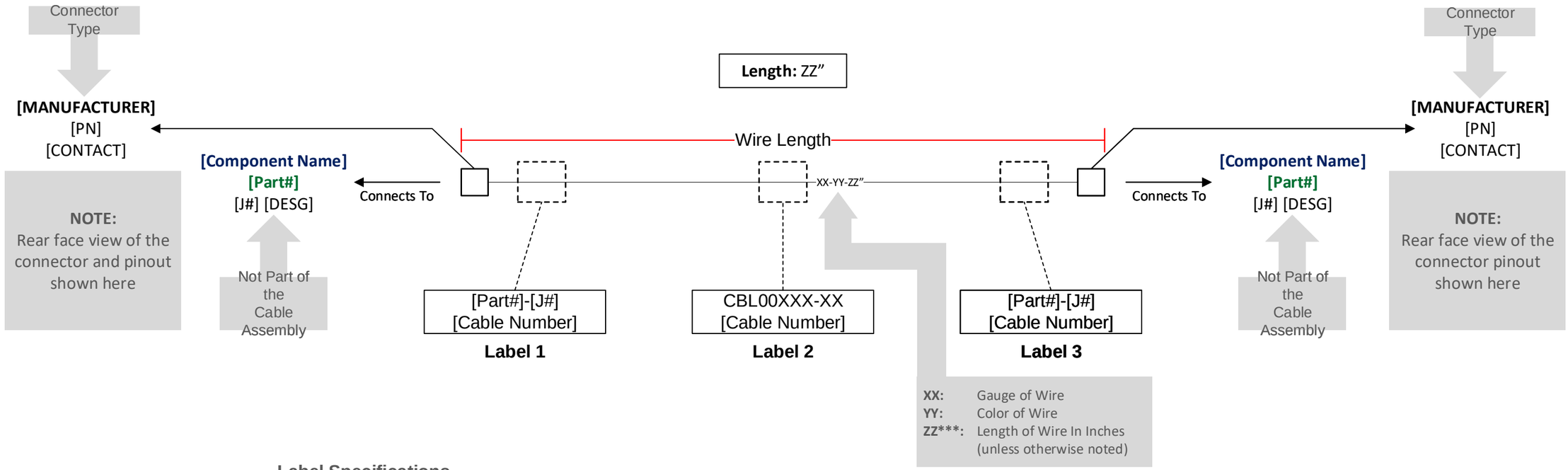
CACI EVER VIGILANT
45200 Business Court
Dulles, VA 20166

TITLE
CABLE ASSEMBLY DIAGRAM, UNIVERSAL YELLOWSTONE CHASSIS

SIZE	CAGE CODE	DRAWING NO	REV
B	1KYV6	CBL00040-01	A

UNCLASSIFIED

Cable Assembly Specifications



XX: Gauge of Wire
YY: Color of Wire
ZZ*:** Length of Wire In Inches (unless otherwise noted)

Label Specifications

Unless otherwise noted, following specifications must be followed.
**If cable length is 6" or less, only add one label (Label 2) in the center.

Type: Heat Shrink (White)
Text Color: Black
Background Color: White
Font Size: As Needed
Side Margin: .1"

Label Location (unless otherwise noted)**:
Label 1: 1" from the end
Label 2: In the Center
Label 3: 1” from the end

Length Specifications

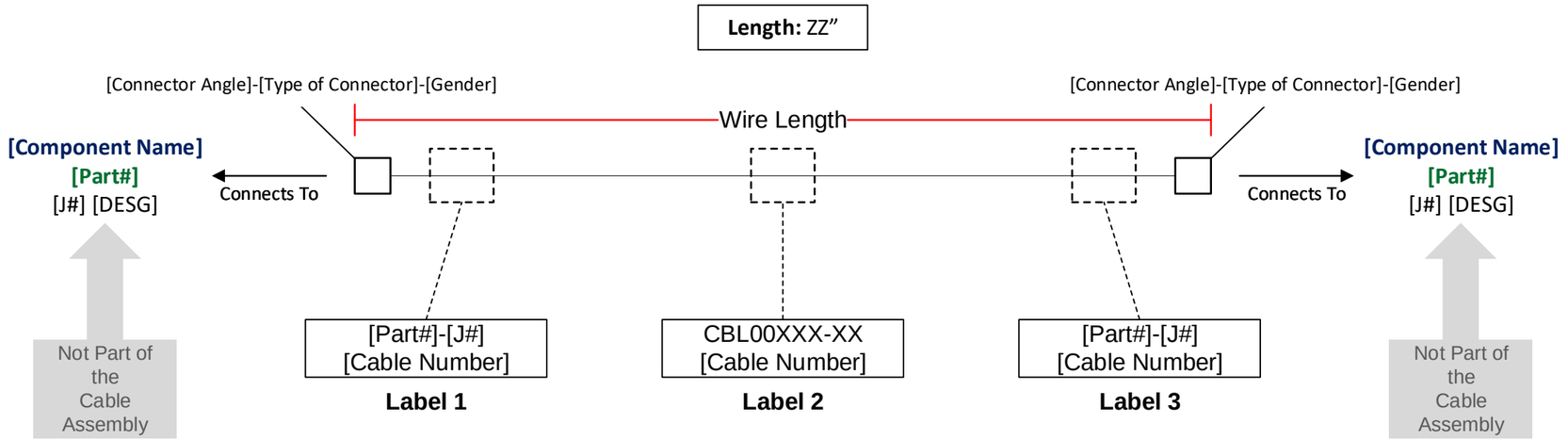
- Length of the wire is from the tip of one connector to the tip of the other connector.
- Cable lengths under 6" can have a tolerance of +/- 0.25"
- Cable lengths above 6" can have a tolerance of +/- 0.5"

FOR PRODUCTION TEAM: Heat shrink **MUST** be used to cover the solder connections.

Example of Bill Of Materials

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1		(P/C/M)XXXX	CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
2			CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
3			CONTACT, Description	Manufacturer Part Number	Manufacturer Name		each
4			WIRE, Description	Generic	Generic		inch
5			WIRE, Description	Generic	Generic		inch

RF Cable Assembly Specifications



Label Specifications

Unless otherwise noted, following specifications must be followed.
**If cable length is 6" or less, only add one label (Label 2) in the center.

Type:	Heat Shrink (White)	**Label Location** (unless otherwise noted)**:	
Text Color:	Black	**Label 1:**	1" from the end
Background Color:	White	**Label 2:**	In the Center
Font Size:	As Needed	**Label 3:**	1” from the end
Side Margin:	.1"		

Length Specifications

- Length of the wire is from the tip of one connector to the tip of the other connector.
- Cable lengths under 6" can have a tolerance of +/- 0.1"
- Cable lengths above 6" can have a tolerance of +/- 0.25"

FOR PRODUCTION TEAM: Heat shrink **MUST** be used to cover the solder connections.

Example of Bill Of Materials

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1		RXXXX	CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
2			CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
4			CABLE, Description	Generic	Generic		inch

SIZE	CAGE CODE	DRAWING NO	REV
B	1KYV6	CBL00040-01	A

Sheet 3 of 12

10 MHz IN

R0001

C1



C2



Length: 6"

RA-SMA-M

SMA Bulkhead
SM4939
10 MHz IN

Connects To

RA-SMA-M

Connects To

Chassis RF/Power Dist. PCB
PCB00064-01
J8

CBL00040-01
R0001

Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1-2	R0001	CONNECTOR, SMA Connector Plug, Male Pin, Right Angle	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inches

10 MHz OUT

R0002

C1



C2



Length: 6"

RA-SMA-M

SMA Bulkhead
SM4939
10 MHz OUT

Connects To

RA-SMA-M

Connects To

Chassis RF/Power Dist. PCB
PCB00064-01
J7

CBL00040-01
R0002

Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1-2	R0002	CONNECTOR, SMA Connector Plug, Male Pin, Right Angle	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inches

1 PPS IN

R0003

C1



C2



Length: 6"

RA-SMA-M

SMA Bulkhead
SM4939
1 PPS IN

Connects To

RA-SMA-M

Connects To

Chassis RF/Power Dist. PCB
PCB00064-01
J3

CBL00040-01
R0003

Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1-2	R0003	CONNECTOR, SMA Connector Plug, Male Pin, Right Angle	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inches

1 PPS OUT

R0004

C1



C2



Length: 6"

RA-SMA-M

SMA Bulkhead
SM4939
1 PPS OUT

Connects To

RA-SMA-M

Connects To

Chassis RF/Power Dist. PCB
PCB00064-01
J1

CBL00040-01
R0004

Label 2

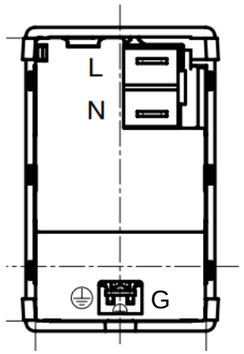
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1	C1-2	R0004	CONNECTOR, SMA Connector Plug, Male Pin, Right Angle	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inches

Module, AC Power Entry to MOV Board

P0001

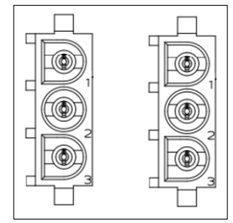
FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

Module, AC Power Entry

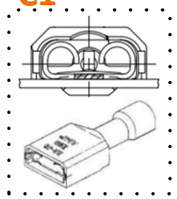


FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

MOV Board

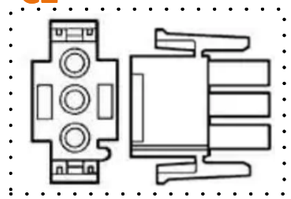


C1



x3

C2



Length: 9"

Module, AC Power Entry
4301.1401, 0034.3126, KMF1.1191.11
J1

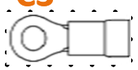
Connects To

L — +120V AC — 16-BK-9" — +120V AC — 1
G — GND — GND — 2
N — RTN — 16-WT-9" — RTN — 3

Connects To

MOV Board
PCB00075
J1

C3



Chassis Ground Screw
ASY00478

16-GN/YL-7"

Chassis Ground
P0001
Label 3B

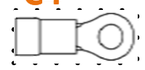
AC Power Entry
P0001
Label 1

CBL00040-01
P0001
Label 2

MOV Board
P0001
Label 3A

16-GN/YL-7"

C4



Connects To

Chassis Ground Screw
ASY00478

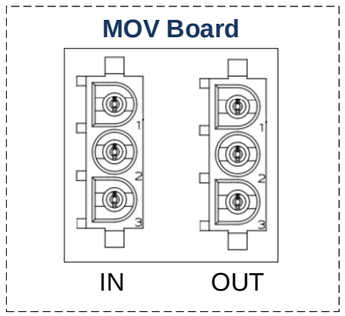
Chassis Ground
P0001
Label 3C

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1	P0001	CONNECTOR, Quick Connect, RCPT 16-14AWG 0.250	3-350819-2	TE Connectivity AMP Connectors	3	each
2	C2		CONNECTOR, Plug 3Pos 94V-0 UNI-MATE	350766-1	TE Connectivity AMP Connectors	1	each
3	"		CONTACT, Pin, 14-20AWG, Crimp, Gold	350687-7	TE Connectivity AMP Connectors	3	each
4	C3-C4		CONNECTOR, Circular 16-14AWG #10 Crimp	7113K24	McMaster-Carr	2	each
5			WIRE, Black, 16AWG	See specifications	-	9	inches
6			WIRE, White, 16AWG	See specifications	-	9	inches
7			WIRE, Green and Yellow, 16AWG	See specifications	-	14	inches

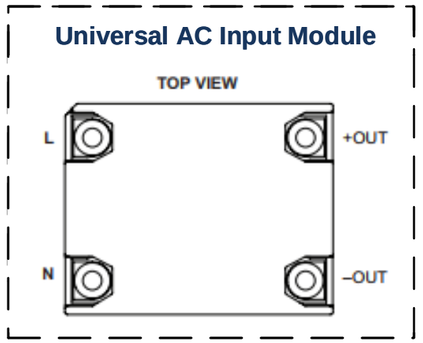
MOV Board to Universal AC Input Module

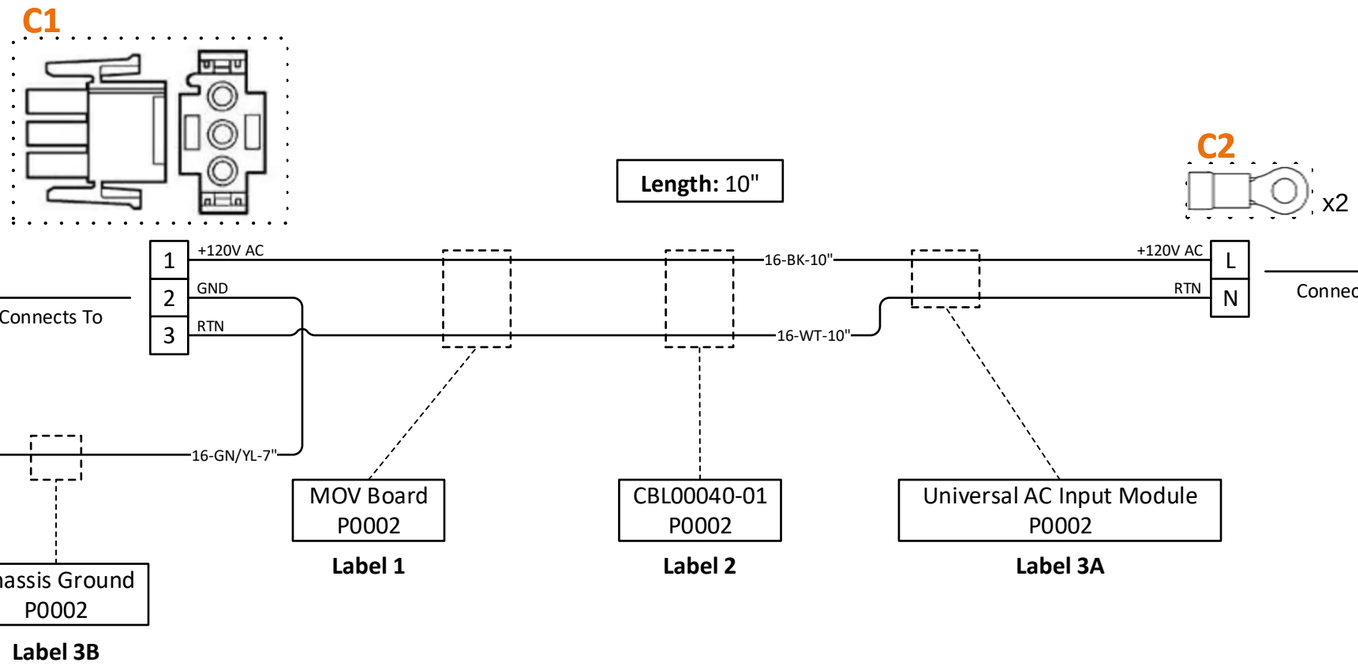
P0002

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY



FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

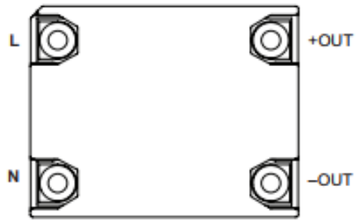




Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1	P0002	CONNECTOR, Plug, 3Pos 94V-0 UNI-MATE	350766-1	TE Connectivity AMP Connectors	1	each
2	"		CONTACT, Pin, 14-20AWG Crimp Gold	350687-7	TE Connectivity AMP Connectors	3	each
3	C2		CONNECTOR, Circular 16-14AWG #6 Crimp	7113K23	McMaster-Carr	2	each
4	C3		CONNECTOR, Circuilar 16-14AWG #10 Crimp	7113K24	McMaster-Carr	1	each
5			WIRE, Black, 16AWG	See specifications	-	10	inches
6			WIRE, White, 16AWG	See specifications	-	10	inches
7			WIRE, Green and Yellow, 16AWG	See specifications	-	7	inches

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

Universal AC Input Module

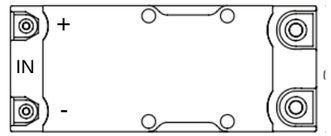


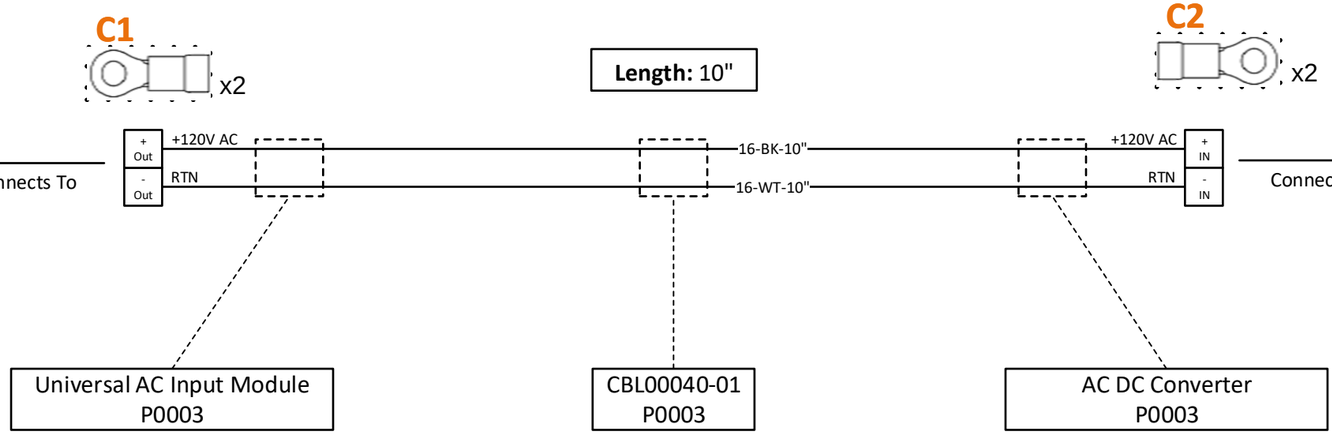
Universal AC Input Module to AC DC Converter

P0003

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

AC DC Converter





Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1-C2	P0003	CONNECTOR, Circular 16-14AWG #6 Crimp	7113K23	McMaster-Carr	4	each
2			WIRE, Black, 16AWG	See specifications	-	10	inches
3			WIRE, White, 16AWG	See specifications	-	10	inches

SIZE	CAGE CODE	DRAWING NO	REV
B	1KYV6	CBL00040-01	A

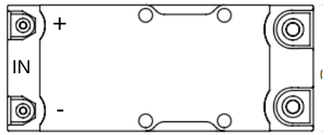
Sheet 10 of 12

AC DC Converter to Capacitors & Chassis RF/Power Dist. PCB

P0004

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

AC DC Converter



FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

Capacitor 1 & 2



Capacitor 1
ALS70A163DE063

Connects To

Length: 7"

Capacitor 2
ALS70A243DF063

Connects To

Length: 7"

Length: 7"

C4



AC DC Converter
PFM4414VB6M24D0CA0

+24V DC
RTN

16-BL-7"
16-BK-7"

C1 x2

CBL00040-01
P0004
Label 2A

AC DC Converter
P0004
Label 1A

Capacitor 1
P0004
Label 3A

C2 x2

CBL00040-01
P0004
Label 2B

Capacitor 1
P0004
Label 1B

Capacitor 2
P0004
Label 3B

C3 x2

CBL00040-01
P0004
Label 2C

Capacitor 2
P0004
Label 1C

Chassis RF/Power Dist. PCB
P0004
Label 3C

1
2
3
4

Chassis RF/Power Dist. PCB
PCB00064-01
VIN

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1	P0004	CONNECTOR, Circular 16-14AWG #6 Crimp	7113K23	McMaster-Carr	2	each
2	C2-C3		CONNECTOR, Circular 16-14AWG #10 Crimp	7113K24	McMaster-Carr	4	each
3	C4		CONTACT, Socket, 14-16AWG, Crimp, Gold	1720630311	Molex	4	each
4	"		CONNECTOR, Receptacle Housing, 4Pos 5.70mm	1716920104	Molex	1	each
5			WIRE, Blue, 16AWG	See specifications	-	21	inches
6			WIRE, Black, 16AWG	See specifications	-	21	inches

SIZE	CAGE CODE	DRAWING NO	REV
B	1KYV6	CBL00040-01	A

Sheet 11 of 12

UNCLASSIFIED

Vin to Fans & LED

P0005

C1



2
1

Rear View

Length: Various

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

*Fan wires are included with the fans

Chassis RF/Power Dist. PCB
PCB00064-01
J5

Connects To

1 +24V DC
2 RTN

22-RD-9"
22-BK-9"

+24V +
GND -

Fan 1
109P0424J3023



+24V +
GND -

Fan 2
109P0424J3023

22-RD-15"
22-BK-15"

+24V +
GND -

Solders to

LED 24V Green
6091222134F

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY



Chassis RF/Power Dist. PCB – J5
P0005

Label 1

CBL00040-01
P0005

Label 2

LED 24V Green
P0005

Label 3

Solder splice and heat shrink

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1	P0005	CONNECTOR, Receptacle Housing, 2Pos, 3mm	WM1783-ND	Molex	1	each
2	"		CONTACT, Socket, 20-24AWG, Crimp, Gold	WM1839-ND	Molex	2	each
3			WIRE, Red, 22AWG	See specifications	-	24	inches
4			WIRE, Black, 22AWG	See specifications	-	24	inches

SIZE	CAGE CODE	DRAWING NO	REV
B	1KYV6	CBL00040-01	A

Sheet 12 of 12

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